

Math 242, S20
Exam 1

Name: *Answer*

Instructions:

- There are totally 5 problems.
- You must show all of your work to receive a credit for a correct answer.

Problem	Points	Score
1	20	
2	22	
3	22	
4	18	
5	18	
Total	100	

Good Luck !

Problem 1. (20 points) Find the particular solution of the following initial value problems:

(a) $\frac{dy}{dx} = \frac{-3}{\sqrt{2x+1}}, \quad y(4) = -1.$

$$y = \int \frac{-3}{\sqrt{2x+1}} dx = -3\sqrt{2x+1} + C$$

$$y(4) = -1 \Rightarrow -3\sqrt{2 \cdot 4 + 1} + C = -1$$

$$-3 \cdot 3 + C = -1$$

$$C = -1 + 9 = 8$$

$$\Rightarrow \underline{y(x) = -3\sqrt{2x+1} + 8}$$

(b) $(1+x^2)\frac{dy}{dx} - 2x(1+x^2)y = 2xe^{x^2}, \quad y(0) = 2.$

$$\frac{dy}{dx} - 2xy = \frac{2xe^{x^2}}{1+x^2} \leftarrow \text{standard form}$$

$$P(x) = -2x, \quad Q(x) = \frac{2xe^{x^2}}{1+x^2}$$

$$\text{integrating factor: } \rho(x) = e^{\int P(x) dx} = e^{\int -2x dx} = e^{-x^2}$$

Thus

$$\frac{d}{dx}[ye^{-x^2}] = \frac{2xe^{x^2}}{1+x^2} \cdot e^{-x^2}$$

$$ye^{-x^2} = \int \frac{2x}{1+x^2} dx$$

$$ye^{-x^2} = \ln(1+x^2) + C$$

$$y = e^{x^2} [\ln(1+x^2) + C]$$

$$y(0) = 2 \Rightarrow 2 = e^{0^2} [\ln(1+0^2) + C] \Rightarrow C = 2$$

$$\Rightarrow \underline{y(x) = e^{x^2} [\ln(1+x^2) + 2]}$$

Problem 2. (22 points) Find a general solution for the following differential equations.

(a) $\frac{dy}{dx} = 1 + x^2 + y + x^2y.$ (Hint: Factor the right-hand side.)

$$\frac{dy}{dx} = (1+x^2)(1+y)$$

$$\frac{1}{1+y} dy = (1+x^2) dx$$

$$\ln|1+y| = x + \frac{x^3}{3} + C$$

$$|1+y| = e^{x + \frac{x^3}{3} + C}$$

$$1+y = A e^{x + \frac{x^3}{3}}$$

$$\underline{y = A e^{x + \frac{x^3}{3}} - 1}$$

(b) $(2xy - \sin x) dx + (x^2 - \cos y) dy = 0$

$$M(x,y) = 2xy - \sin x \Rightarrow \frac{\partial M}{\partial y} = 2x \Rightarrow \text{exact}$$

$$N(x,y) = x^2 - \cos y \Rightarrow \frac{\partial N}{\partial x} = 2x$$

$$F(x,y) = \int M(x,y) dx = \int (2xy - \sin x) dx$$

$$= x^2y + \cos x + g(y)$$

$$\frac{\partial F}{\partial y} = N \Rightarrow \frac{\partial}{\partial y} [x^2y + \cos x + g(y)] = x^2 - \cos y$$

$$x^2 + g'(y) = x^2 - \cos y$$

$$g'(y) = -\cos y$$

$$\Rightarrow g(y) = \int -\cos y dy = -\sin y + C$$

$$\text{Thus } F(x,y) = x^2y + \cos x - \sin y + C$$

and the solution is given by

$$\underline{x^2y + \cos x - \sin y = C}$$

Problem 3. (22 points) Find a general solution for the following differential equations by using appropriate substitutions.

(a) $\frac{dy}{dx} = (x + 3y)^2$

Let $v = x + 3y$, then $y = \frac{v-x}{3}$ and $\frac{dy}{dx} = \frac{1}{3} \frac{dv}{dx} - \frac{1}{3}$

thus we get

$$\frac{1}{3} \frac{dv}{dx} - \frac{1}{3} = v^2$$

$$\frac{dv}{dx} = 1 + 3v^2$$

$$\frac{1}{1+3v^2} dv = dx$$

$$\int \frac{1}{1+3v^2} dv = \int dx$$

$$\frac{1}{\sqrt{3}} \arctan(\sqrt{3}v) = x + C$$

$$\arctan(\sqrt{3}v) = \sqrt{3}x + C$$

$$v = \frac{1}{\sqrt{3}} \tan(\sqrt{3}x + C)$$

$$\Rightarrow x + 3y = \frac{1}{\sqrt{3}} \tan(\sqrt{3}x + C)$$

$$3y = \frac{1}{\sqrt{3}} \tan(\sqrt{3}x + C) - x$$

$$\underline{y(x) = \frac{1}{3\sqrt{3}} \tan(\sqrt{3}x + C) - \frac{x}{3}}$$

(b) $2xyy' = x^2 + 2y^2$

$$y' = \frac{x^2 + 2y^2}{2xy} \Rightarrow y' = \frac{1}{2} \left(\frac{x}{y} \right) + \frac{y}{x}$$

Let $v = \frac{y}{x}$, then $y = vx$ and $\frac{dy}{dx} = x \frac{dv}{dx} + v$

Thus we get

$$x \frac{dv}{dx} + v = \frac{1}{2v} + v$$

$$x \frac{dv}{dx} = \frac{1}{2v}$$

$$2v dv = \frac{1}{x} dx$$

$$\int 2v dv = \int \frac{1}{x} dx$$

$$v^2 = \ln|x| + C$$

$$\Rightarrow \frac{y^2}{x^2} = \ln|x| + C$$

$$y^2 = x^2 (\ln|x| + C)$$

$$\Rightarrow \underline{y(x) = x (\ln|x| + C)^{\frac{1}{2}} \text{ or } y(x) = -x (\ln|x| + C)^{\frac{1}{2}}}$$

Problem 4. (18 points) Suppose that a motorboat is moving at 60 ft/s (feet/second) when its motor suddenly quits at time $t = 0$, and that 10 seconds later the motorboat has slowed to 30 ft/s. Assume that the resistance it encounters while coasting is proportional to its velocity, i.e., $\frac{dv}{dt} = -kv$.

(a) Find the velocity $v(t)$ of the motorboat and determine all parameters in the solution using the given conditions.

$$\frac{dv}{dt} = -kv$$

$$\int \frac{1}{v} dv = \int -k dt$$

$$\ln|v| = -kt + C \Rightarrow |v| = e^{-kt+C}$$

$$\Rightarrow \underline{v(t) = Ae^{-kt}} \quad (A \neq 0)$$

$$v(0) = 60 \Rightarrow A = 60 \Rightarrow v(t) = 60e^{-kt}$$

$$v(10) = 30 \Rightarrow 60e^{-k(10)} = 30 \Rightarrow e^{-10k} = \frac{1}{2}$$

$$\Rightarrow -10k = \ln\left(\frac{1}{2}\right) \Rightarrow k = \frac{1}{10}\ln 2$$

$$\text{Thus } v(t) = 60e^{-(\frac{\ln 2}{10})t} \approx 60e^{-0.0693t}$$

(b) How long does it take for the motorboat to slow to 5 ft/s after its motor quits?

$$\begin{aligned} v(t) &= 5 \\ \Rightarrow 60e^{-(\frac{\ln 2}{10})t} &= 5 \\ e^{-(\frac{\ln 2}{10})t} &= \frac{5}{60} = \frac{1}{12} \end{aligned}$$

$$-(\frac{\ln 2}{10})t = -\ln 12$$

$$t = 10 \cdot \frac{\ln 12}{\ln 2} \approx 35.8 \text{ seconds}$$

Problem 5. (18 points) Given the following initial value problem:

$$\frac{dx}{dt} = 3x^2 - 18x, \quad x(0) = 12$$

(a) Find all critical points of the given autonomous differential equation and then determine whether it is stable or unstable for each of them.

$$f(x) = 3x^2 - 18x = 3x(x-6) = 0$$

$\Rightarrow x=0, x=6 \rightarrow$ two critical points

$$\begin{array}{ccc} x' > 0 & x' < 0 & x' > 0 \\ \rightarrow & \leftarrow & \rightarrow \\ 0 & 6 & \end{array} \quad \begin{array}{l} x=0 \text{ is stable} \\ x=6 \text{ is unstable} \end{array}$$

(b) Solve the given initial value problem for $x(t)$ and describe the behavior of $x(t)$ when t increases from 0.

$$\frac{dx}{dt} = 3x(x-6)$$

$$\int \frac{1}{x(x-6)} dx = \int 3 dt$$

$$\int \frac{1}{6} \left(\frac{1}{x-6} - \frac{1}{x} \right) dx = \int 3 dt$$

$$\int \frac{1}{x-6} - \frac{1}{x} dx = \int 18 dt$$

$$\ln|x-6| - \ln|x| = 18t + C$$

$$\left| \frac{x-6}{x} \right| = e^{18t+C}$$

$$\frac{x-6}{x} = A e^{18t} \quad (A \neq 0)$$

$$x(0)=12 \Rightarrow \frac{12-6}{12} = A$$

$$A = \frac{1}{2}$$

thus

$$\frac{x-6}{x} = \frac{1}{2} e^{18t}$$

$$1 - \frac{6}{x} = \frac{1}{2} e^{18t}$$

$$\frac{6}{x} = 1 - \frac{1}{2} e^{18t}$$

$$x(t) = \frac{6}{1 - \frac{1}{2} e^{18t}} \Rightarrow x(t) = \frac{12}{2 - e^{18t}}$$

$$2 - e^{18t^*} = 0 \Leftrightarrow e^{18t^*} = 2$$

$$18t^* = \ln 2$$

$$t^* = \frac{1}{18} \ln 2$$

When $t \rightarrow t^*-$, we have

$x(t)$ increase to $+\infty$, i.e.

$$\lim_{t \rightarrow t^*-} x(t) = \lim_{t \rightarrow t^*-} \frac{12}{2 - e^{18t}} = +\infty$$